a). Design a circuit to verify Kirchoff's Voltage law for input voltages V=8V. The circuit uses R1= 1kΩ and R2= 2 kΩ. This setup is connected across a constant voltage source. Also, verify the results practical and theoretical values

a). Design an electrical circuit to verify Kirchoff's current law for three different input voltages which have two resistors R1= 560 ohm and R2= 560 ohm connected in parallel and this combination is connected across a constant DC source. Also, verify the results by current division rule.

a). Analyze the given circuit where a resistor R1=5 kΩ, R2=10kΩ, and R3=7.5 kΩ are connected in a series circuit powered by a 15 V DC source

a). Analyze the given circuit where resistors R1=1kΩ, R2=3kΩ, and R3=2kΩ are connected in parallel to a 6V DC source.

a). A 12V voltage source is connected in series with two resistors, R1=4 Ω and R2=6 Ω. The load resistor RL=8 Ω is connected across the terminals where Thevenin's equivalent is to be found. Verify the results theoretically and practically.

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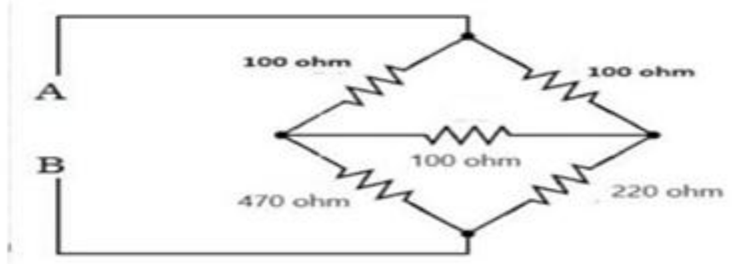
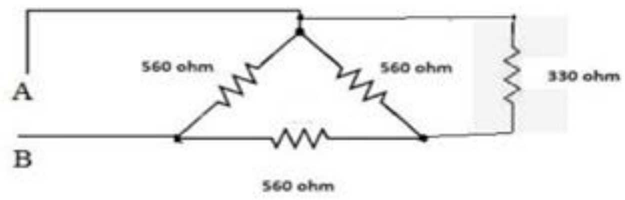
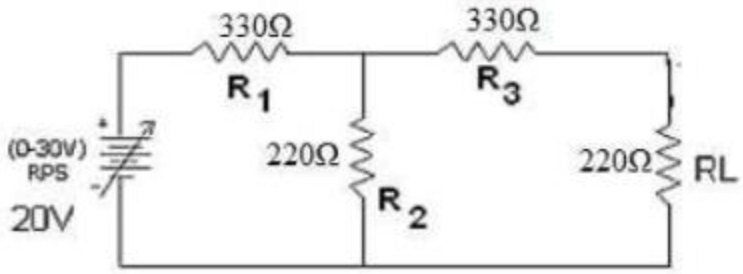
a). Design a circuit to verify star delta reduction technique for the bridge circuit shown below. Calculate the effective resistance between the terminals A & B theoretically and practically.



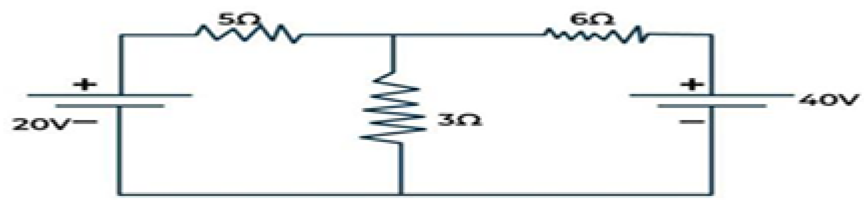
a). Design a circuit to verify star delta reduction technique for the circuit shown below. Calculate the effective resistance between the terminals A & B theoretically and practically



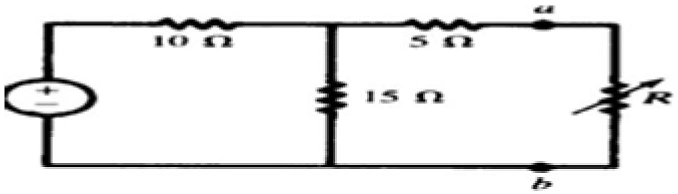
a). Determine the current flows through 330 Ω (R3) resistor by applying Thevenin's & Norton's theorem theoretically and verify the results theoretically and practically. Assume 12V supply.



a). Determine the current flows through 3 Ω resistor by applying Superposition theorem theoretically and verify the results theoretically and practically. Assume 20V supply.



a). Find the value of the adjustable resistance R that dissipates the maximum power across terminals a-b. What is the maximum power that can be delivered to this load? Assume supply 15V.



a) Design and determine the output characteristics of LVDT and calibrate the measuring instruments for given frequency and no of turns.

a). Implement the domestic wiring used to control a single electrical load from two different places by using 2-way switches and a bulb. Also verify its operation using truth table.

S.NO	S1	S2	LAMP STATUS
1	CA	C'A'	ON
2	CB	C'B'	ON
3	CA	C'B'	OFF
4	CB	C'A'	OFF